



The problem

Epigenomic data form a three-dimensional tensor across assays, cell types and genomic positions, with tens to hundreds of assays and thousands of cell types. Due to the high cost and complexity of profiling every possible assay in all relevant cell and tissue types, this tensor is sparse: most experiments were never performed. Epigenome imputation methods aim to predict these missing experiments [1].

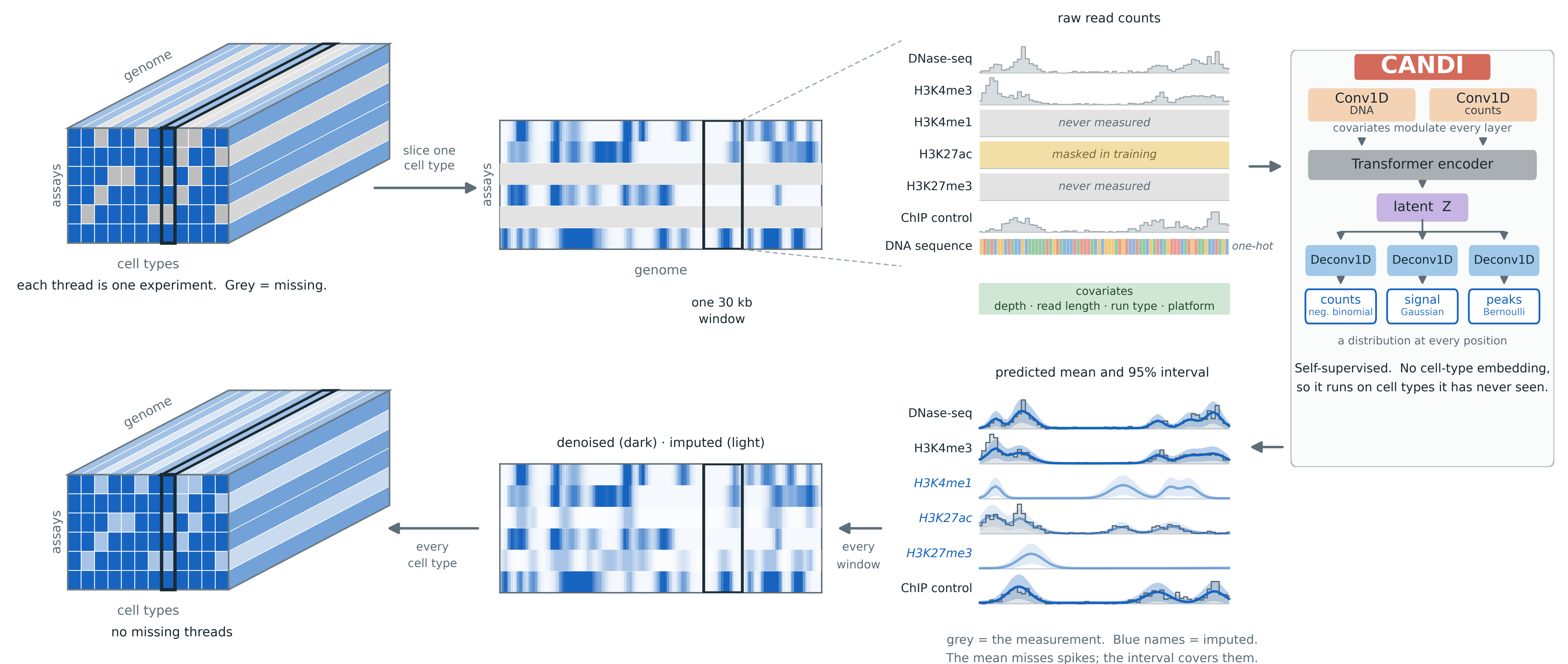
CANDI

Confidence-Aware Neural Denoising Imputer [2] is a self-supervised deep learning method for epigenome imputation and denoising. It takes raw read counts, DNA sequence and experimental covariates, and outputs probability distributions over counts, signal and peaks. Conditioning on metadata rather than cell-type embeddings enables zero-shot generalization to unseen cell types.

Key findings

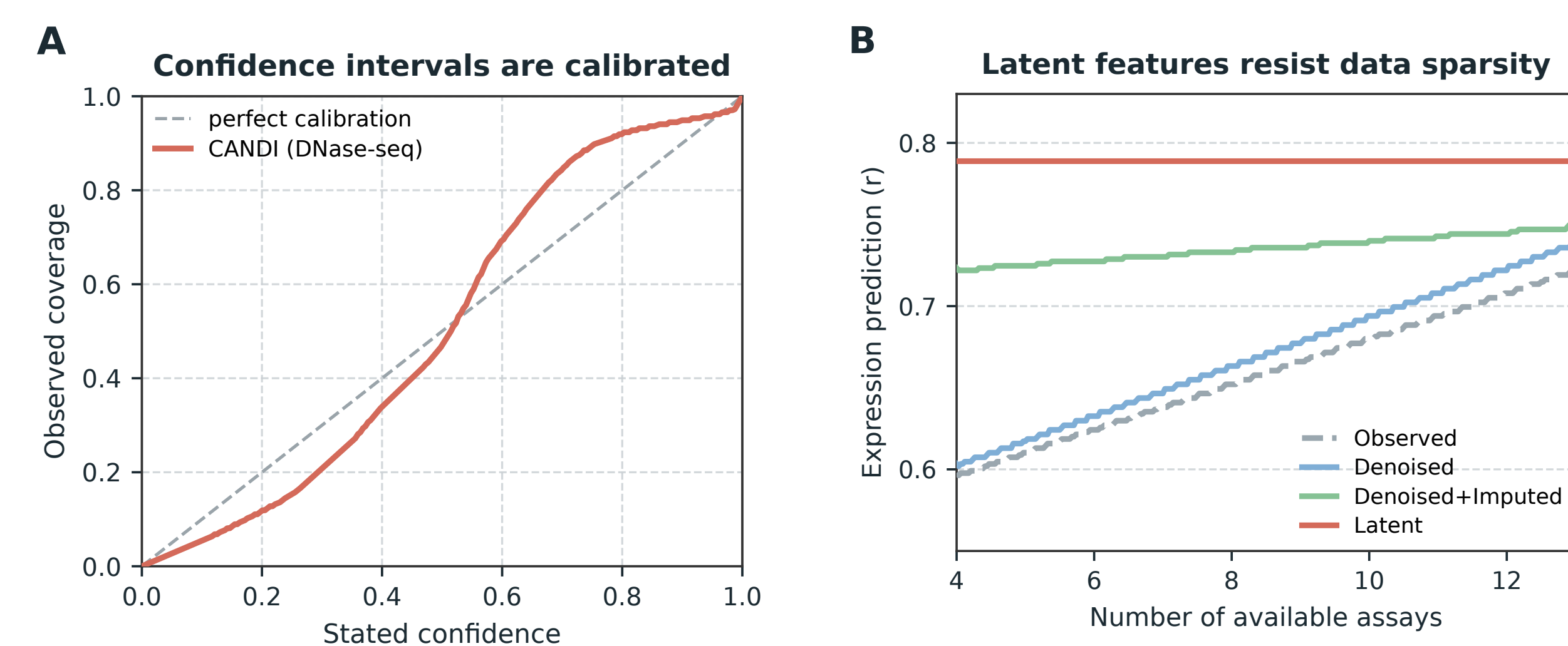
- CANDI ranks **1st of 26** on the ENCODE Imputation Challenge.
- CANDI provides calibrated uncertainty estimates.
- CANDI's imputed signals and latent space predict gene expression.

CANDI schematic workflow



Top: for each 30 kb locus, CANDI takes observed read counts with the ChIP-seq control, four covariates per assay and the DNA sequence; grey assays were never measured, yellow ones are masked. **Bottom:** each output is a probability distribution, here a mean and 95% interval per assay; window by window, they fill every missing thread of the tensor.

Downstream utility of CANDI



A Confidence calibration for DNase-seq (DND-41): predicted confidence level against the fraction of observed data within that interval.

B Gene expression prediction as input assays are removed. Denoised+Imputed and Latent settings are robust to input sparsity.

[1] Schreiber J, Boix C, *et al.* The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

[2] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels C-D: blind-test scoring from the Libbrecht lab's EIC reproduction; *baseline* is the challenge's average-activity track. The correction was chosen after the challenge results were public; none of it is fitted on blind-test data.

Fixing the train-test experimental covariate shift in the ENCODE Imputation Challenge

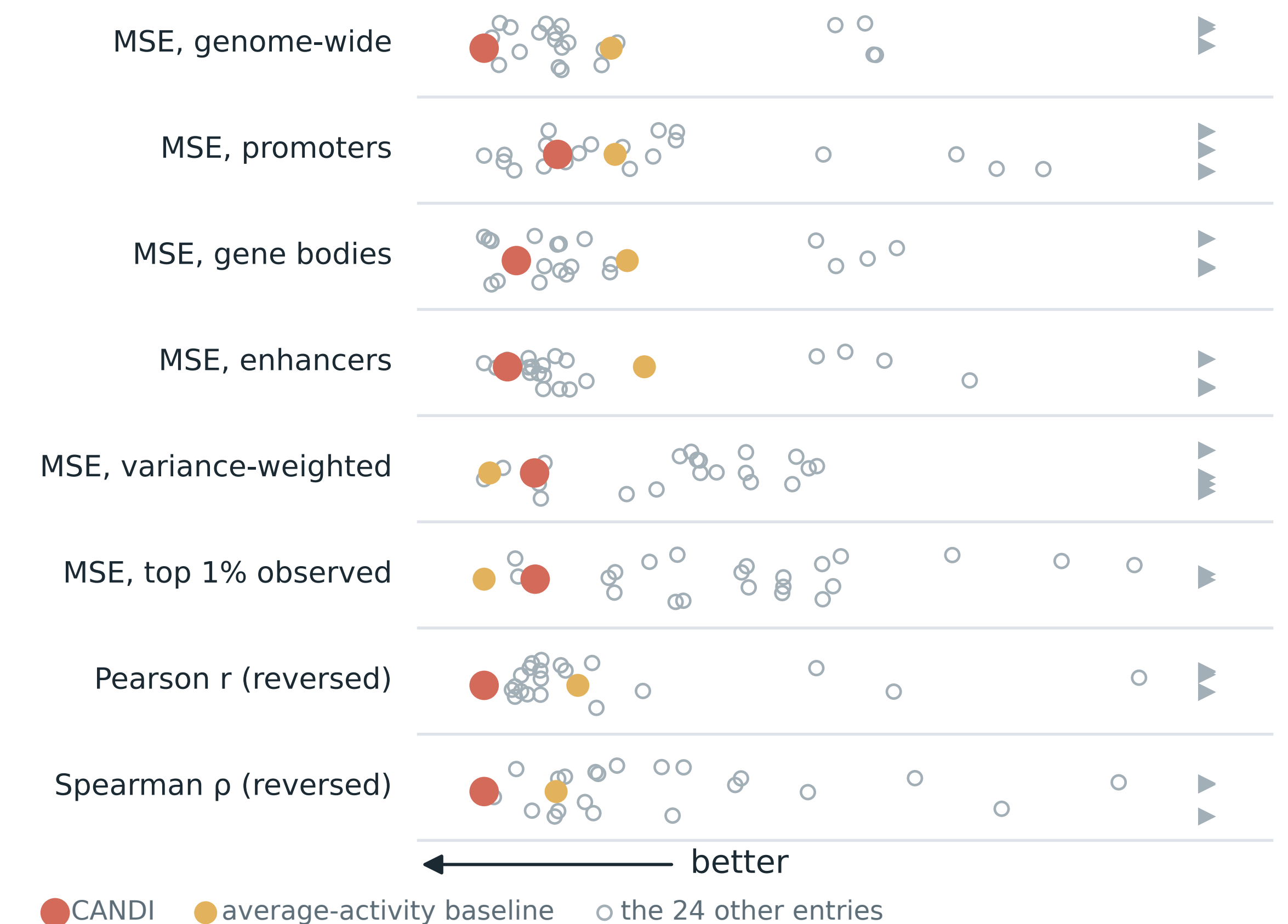
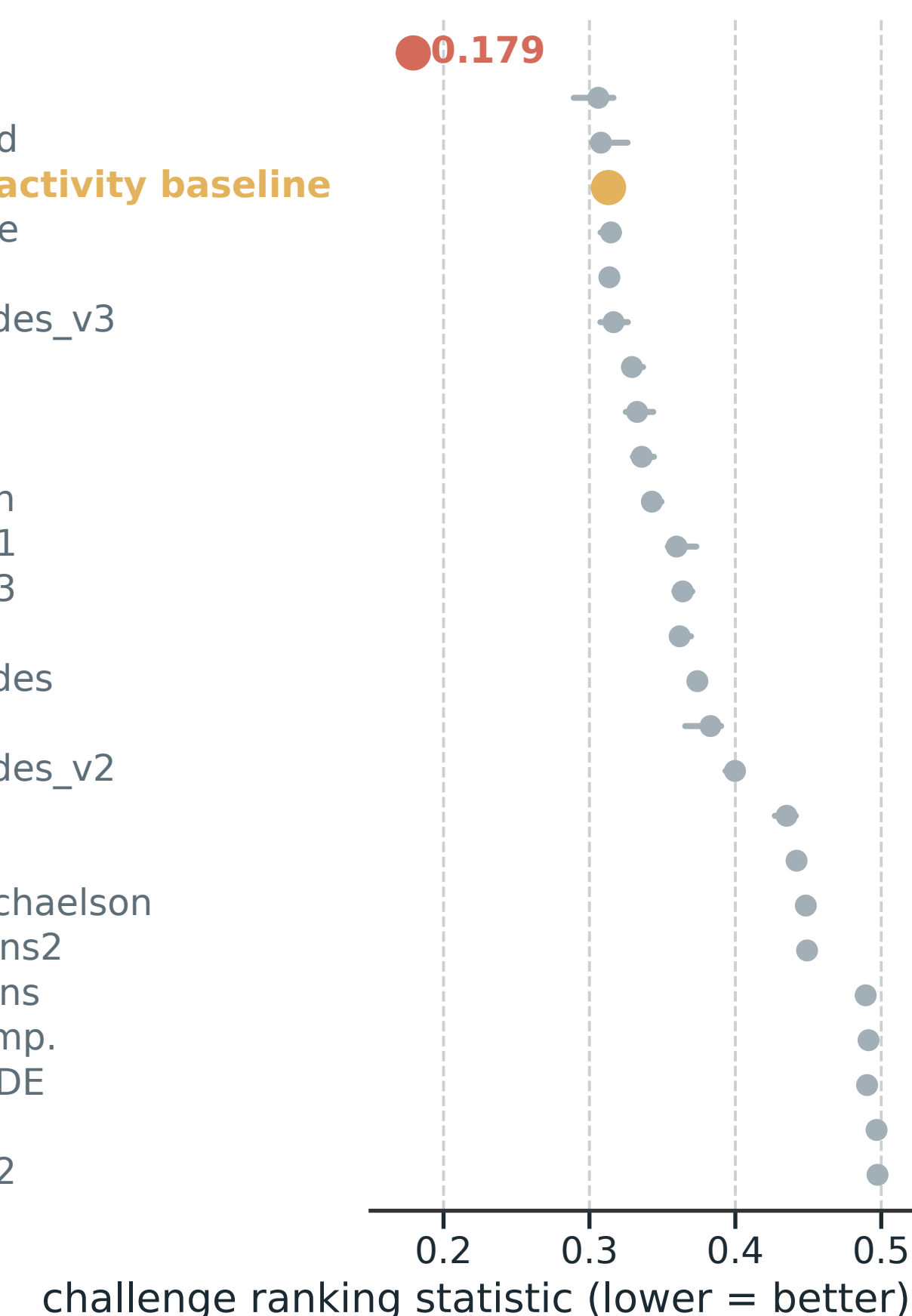
The shift. We followed the EIC train-test split: 51 blind-test experiments (12 cell types, 8 assays). The train and test sets differ subtly: every blind-test experiment is paired-end, against 15% of training experiments, and blind-test tracks sit at an assay-specific signal level. CANDI recovers the rank-ordering and structure of the signal but not its absolute magnitude; its raw output ranks 14th.

The fix. Per assay, CANDI's output passes through one layer,

$$\max(\beta_0 + \beta_1 \cdot \text{CANDI} + \beta_2 \cdot \text{baseline}, 0),$$

fitted by least squares on the 133 training experiments only. CANDI then ranks 1st.

- 1 CANDI
- 2 HLYG v1
- 3 Lavawizard
- 4 average-activity baseline
- 5 Guacamole
- 6 imp
- 7 BrokenNodes_v3
- 8 imp1
- 9 HLYG
- 10 CUWA
- 11 LiPingChun
- 12 CUImpute1
- 13 Song Lab 3
- 14 ICU
- 15 BrokenNodes
- 16 Song Lab
- 17 BrokenNodes_v2
- 18 CostaLab
- 19 HLYG v2
- 20 UIOWA Michaelson
- 21 NittanyLions2
- 22 NittanyLions
- 23 Aug2019Imp.
- 24 KKT-ENCODE
- 25 Avocado
- 26 Song Lab 2



C The leaderboard with CANDI added: CANDI is first at 0.179 (bootstrap range 0.177–0.184), the next entry 0.306.

D Eight of the nine measures relative to the baseline, median over 51 experiments. CANDI is first on three and top 8 on the rest.